

MAT1110 - Foundation Mathematics and Statistics for Social Sciences

Pre-requisite: 'O' Level Mathematics

Course Content

1. Sets

Definition; subsets; intersection and union; Venn diagrams.

2. Numbers

Natural numbers; integers, rational numbers; Real numbers; complex numbers.

3. Functions

Relations; domain, range; graphs, one functions, inverse functions.

4. Polynomial Functions

Linear; quadratic; quadratic equations; completing the square, polynomials; the Factor theorem; the Remainder theorem; synthetic division; rational functions; partial fractions, inequalities.

5. Trigonometry

Definition of trigonometric functions; identities; equations; graphs.

6. Exponential and Logarithmic Functions

Definition; properties; graphs; equations.

7. Differential Calculus

Limits and continuity; derivatives; rules for differentiation of sums, products, quotients and composite functions; derivatives of standard functions, higher order derivatives; applications to gradients, increasing and decreasing functions, local extreme points; curve sketching.

8. Integral Calculus

The indefinite integral as the inverse differential operation; integration of simple standard forms ($e^x, a^x, \sin x, \cos x, f'(x)/f(x)$), basic methods of integration (substitution, by parts and partial fractions).

9. Descriptive Statistics

Discrete and continuous data; frequency distribution; histograms and frequency polygons; cumulative frequency curves; measures of central tendency (including use of assumed mean); measures of dispersion.

10. Probability

Sample space; events, definition of probability; mutually exclusive events; conditional probability; independent events; Bayes theorem; Probability trees.

Mode of delivery: 3 lectures and 1 tutorial per week.

Assessment

1.	Continuous Assessment	30%
	1.1 Assignments/Quizzes	10%
	1.2 Tests	20%
2.	Final Examination	70%
	Total	100%

Prescribed Readings

Larson R. and Hodgkins A., (2013) College Algebra and Calculus; 2nd edition, Brooks/Cole Cengage Learning, ISBN : 1 133 10518 3

Recommended Readings

Kaufmann J.E., (1994) College Algebra and Trigonometry, PWS Publishers. ISBN : 0 534 93525 2